

MODULE 19 · DOMAIN 4 - BILLING, PRICING, AND SUPPORT

Cost Management Tools

Estimating, watching, analysing and alerting. Four questions, four tools.

DOMAIN 4 of 4 EXAM WEIGHT 12% MODULE 19 of 26

Sort the question by **time**. Before you spend, while you spend, or after you spent?

The tools, mapped to time

WHEN	TOOL	WHAT IT DOES
Before	AWS Pricing Calculator	Estimate something not yet built
During	AWS Budgets	Alert before you exceed a threshold
During	Cost Anomaly Detection	Spot unusual spend automatically
After	AWS Cost Explorer	Visualise what was actually spent
After	Cost and Usage Report	Detailed line items for your own tools

ON THE EXAM

Estimating something unbuilt is always the Pricing Calculator. Analysing real spend is always Cost Explorer.

AWS Budgets or Cost Explorer?

AWS BUDGETS

Looks forward

- You set a threshold and get alerted
- Alerts on actual or forecast spend
- Cost, usage, Reserved Instance and Savings Plans budgets
- Acts before you overspend

AWS COST EXPLORER

Looks backward

- Visualise and analyse historical cost and usage
- Filter and group by service, tag, Region or account
- Explains what already happened
- Also forecasts, based on that history

THE TIEBREAKER

Alert me when we exceed means Budgets. Show me what we spent and where means Cost Explorer.

AWS Budgets

- ◆ **Set a threshold for cost or usage** and be notified when actual or forecast spend crosses it.
- ◆ Budgets can be scoped to a service, a tag, a linked account, or the whole organisation.
- ◆ **Forecast-based alerts warn you before the month ends**, which is the point of the service.
- ◆ There are also budgets for Reserved Instance and Savings Plans utilisation and coverage.
- ◆ Budget actions can automatically apply a policy or stop resources when a threshold is breached.

ON THE EXAM

Notify the team before spend exceeds an allocation is Budgets. A CloudWatch billing alarm is the older, cruder approach.

AWS Cost Explorer

- ◆ **Visualises historical cost and usage** with graphs and tables you can filter and group.
- ◆ **Group by linked account** is how a company reports spend per department when each has its own account.
- ◆ It also groups by service, Region, instance type and cost allocation tag.
- ◆ It provides rightsizing recommendations, and Reserved Instance and Savings Plans purchase recommendations.
- ◆ It forecasts future spend, but that forecast is derived from past usage rather than from a plan.

ON THE EXAM

Break the bill down by department, team or project in Cost Explorer, grouping by linked account or by tag.

Cost Explorer or the Cost and Usage Report?

AWS COST EXPLORER

A tool you look at

- Graphs and tables in the console
- Answers questions directly
- Good enough for most reporting
- No setup beyond enabling it

COST AND USAGE REPORT

A file you process

- The most granular billing data available
- Delivered to Amazon S3 on a schedule
- Queried with Athena, Redshift or a BI tool
- For custom analysis beyond what the console offers

THE TIEBREAKER

Console analysis means Cost Explorer. Feeding billing data into your own systems means the Cost and Usage Report.

AWS Pricing Calculator

- ◆ **Estimates the cost of an architecture before you build it.**
- ◆ You describe the services and quantities; it produces a monthly estimate you can share.
- ◆ **It never sees your account.** It has no knowledge of what you actually run or spend.
- ◆ Useful for business cases, comparing architectures, and budgeting a new project.
- ◆ It cannot compare against your current on-premises costs. Migration Evaluator does that.

ON THE EXAM

If nothing has been built yet, it is the Pricing Calculator. Every other tool here needs existing usage.

Cost allocation tags

- ◆ **Tags are key-value labels on resources**, and cost allocation tags make them appear in billing data.
- ◆ This is how a company attributes spend to a project, team or environment within one account.
- ◆ **They must be activated in the billing console** before they appear in reports.
- ◆ Activation is not retrospective. Data before activation is not tagged in the billing record.
- ◆ AWS-generated tags are created by AWS; user-defined tags are ones you apply yourself.

ON THE EXAM

Separate accounts per department means group by linked account. One shared account means cost allocation tags.

Consolidated billing

- ◆ **One bill for all accounts** in an AWS Organization, paid by the management account.
- ◆ **Usage is aggregated across accounts**, so volume discount tiers are reached sooner.
- ◆ Reserved Instance and Savings Plans benefits can apply across accounts in the organisation.
- ◆ Each account's individual costs remain visible, so nothing is hidden by combining them.
- ◆ AWS Organizations itself costs nothing. The saving comes purely from aggregation.

ON THE EXAM

Aggregating usage so volume discounts apply sooner is the most-tested consolidated billing fact.

AWS Cost Anomaly Detection

- ◆ **Uses machine learning to learn your normal spending pattern**, then alerts on deviations from it.
- ◆ **It detects unusual spend, not spend above a threshold.** That distinction decides questions.
- ◆ It catches a service suddenly costing more, or a resource left running by mistake.
- ◆ A department that steadily overspends every month is not anomalous, so it will stay silent.
- ◆ It is free to use, and alerts can be sent by email or through Amazon SNS.

ON THE EXAM

Unexpected or unusual spike means Anomaly Detection. Crossing a planned allocation means Budgets.

Reverse lookup

IF THE SCENARIO SAYS	THE ANSWER IS
Estimate the cost before we build it	AWS Pricing Calculator
Alert us before we exceed our monthly allocation	AWS Budgets
Show last quarter's spend broken down by department	AWS Cost Explorer
Detect an unexpected spike in spending	AWS Cost Anomaly Detection
Load detailed billing line items into our own tools	Cost and Usage Report
Attribute spend to projects within one account	Cost allocation tags
One bill and better volume discounts across accounts	Consolidated billing

ON THE EXAM

Seven phrasings covering nearly every cost management question. Time and intent do all the deciding.

A finance director wants a monthly report of AWS spend broken down by department. Each department has its own AWS account, and all accounts sit under one organisation with consolidated billing.

Separately, the director wants an email whenever any department's forecast monthly spend is on track to exceed its allocation.

Which two services meet these requirements? (Choose two.)

- | | |
|---|---|
| ☐ A AWS Cost Explorer | ☐ B AWS Budgets |
| ☐ C AWS Cost and Usage Report | ☐ D AWS Pricing Calculator |
| ☐ E AWS Cost Anomaly Detection | |

◆ PAUSE AND DECIDE

Which two services meet these requirements? (Choose two.)

- A** AWS Cost Explorer
Analyses historical cost grouped by linked account, which is exactly the departmental breakdown.
- B** AWS Budgets
Sets a threshold per account and alerts on forecast spend crossing it.
- C** AWS Cost and Usage Report
The most detailed data available, but it is a raw file for analysis, not a report or an alert.
- D** AWS Pricing Calculator
Estimates the cost of things not yet built. It never sees actual spend.
- E** AWS Cost Anomaly Detection
Alerts on unusual spikes, not on crossing a planned allocation. Different trigger entirely.

THE SIGNAL

Backward-looking breakdown is Cost Explorer. Forward-looking threshold alert is Budgets. Two requirements, two tools.

A company is planning to migrate a three-tier application to AWS. Before committing, the architecture team needs a monthly cost estimate for the proposed design, covering EC2 instances, an RDS database and S3 storage, to include in a proposal for the board.

Nothing has been deployed to AWS yet.

Which service should the team use?

- ☐ A AWS Cost Explorer
- ☐ B AWS Budgets
- ☐ C AWS Pricing Calculator
- ☐ D AWS Cost and Usage Report

◆ PAUSE AND DECIDE

Which service should the team use?

- A

AWS Cost Explorer
Analyses spend that has already happened. There is no AWS spend to analyse.
- B

AWS Budgets
Monitors spend against a threshold. There is nothing running to monitor.
- C

AWS Pricing Calculator
Estimates the cost of a design before it exists, and produces a shareable estimate.
- D

AWS Cost and Usage Report
Detailed billing data for existing usage, of which there is none.

THE SIGNAL

Nothing has been deployed yet eliminates every tool that reads your account. Only the Pricing Calculator works without usage.

A company runs everything in a single AWS account shared by four project teams. Finance wants to know how much each project costs, but every resource sits in the same account so there is nothing to group by.

The company is willing to change how resources are labelled.

What should the company implement?

- ☐ A Create a separate AWS account for each project
- ☐ B Activate cost allocation tags and tag resources by project
- ☐ C Enable AWS Cost Anomaly Detection
- ☐ D Generate a Cost and Usage Report

◆ PAUSE AND DECIDE

What should the company implement?

- A

 Create a separate AWS account for each project
Would work and is a common design, but it is a large restructuring when the scenario offers labelling instead.
- B

 Activate cost allocation tags and tag resources by project
Tags make project labels appear in billing data, so spend can be grouped within one account.
- C

 Enable AWS Cost Anomaly Detection
Finds unusual spend. It does not attribute normal spend to a project.
- D

 Generate a Cost and Usage Report
More detailed data, but without tags the line items still cannot be attributed to a project.

THE SIGNAL

Willing to change how resources are labelled is the tagging invitation. It is placed there to select option B.

A company's AWS bill is normally stable at around thirty thousand dollars a month. In one week, spending on a single service tripled with no corresponding change in the business. Nobody noticed until the invoice arrived.

The company wants to be told automatically the next time something similar happens, without having to predict what it will be.

Which service should the company enable?

- ☐ A AWS Budgets with a monthly cost threshold
- ☐ B AWS Cost Anomaly Detection
- ☐ C AWS Cost Explorer
- ☐ D AWS Trusted Advisor

◆ PAUSE AND DECIDE

Which service should the company enable?

A AWS Budgets with a monthly cost threshold
The strongest distractor. It would fire eventually, but only once the total crosses a number someone chose in advance.

C AWS Cost Explorer
Would show the spike clearly, but only when someone looks. The requirement is to be told automatically.

B AWS Cost Anomaly Detection
Learns the normal pattern and flags deviations, without anyone predicting which service or what amount.

D AWS Trusted Advisor
Best-practice checks including idle resources, but it does not monitor spending patterns.

THE SIGNAL

Without having to predict what it will be rules out thresholds. A threshold is a prediction by definition.

A holding company owns twelve businesses, each with its own AWS account. Each currently pays its own AWS bill separately. Finance wants a single invoice, and has been told the group might also pay less overall.

The businesses must still be able to see their individual costs.

What should the holding company do, and why might costs fall?

- ☐ A Merge all workloads into one AWS account to simplify billing
- ☐ B Use AWS Organizations with consolidated billing; usage aggregates for volume discounts
- ☐ C Use AWS Budgets to combine the twelve bills into one
- ☐ D Use the Cost and Usage Report to produce a single invoice

◆ PAUSE AND DECIDE

What should the holding company do, and why might costs fall?

- A

 Merge all workloads into one AWS account to simplify billing
Destroys the isolation between businesses and is a huge migration. Consolidated billing achieves the goal without it.
- B

 Use AWS Organizations with consolidated billing; usage aggregates for volume discounts
One invoice, individual costs still visible, and aggregated usage reaches volume tiers sooner.
- C

 Use AWS Budgets to combine the twelve bills into one
Budgets alerts on thresholds. It does not produce or combine invoices.
- D

 Use the Cost and Usage Report to produce a single invoice
A data file about usage, not an invoicing mechanism.

THE SIGNAL

One invoice, individual visibility, and lower total. Consolidated billing delivers all three without merging anything.

What to carry out of this module

- ◆ **Sort by time.** Before means Pricing Calculator, during means Budgets, after means Cost Explorer.
- ◆ Nothing deployed yet eliminates every tool except the Pricing Calculator.
- ◆ **Budgets alerts on a threshold you chose. Anomaly Detection alerts on what does not fit the pattern.**
- ◆ Separate accounts means group by linked account. One shared account means cost allocation tags.
- ◆ Cost allocation tags are not retrospective. Activating them does not tag past billing data.

ON THE EXAM

Consolidated billing aggregates usage so volume discounts arrive sooner, while individual account costs stay visible.